

CEN 481 Introduction to Data Mining

GETTING TO KNOW YOUR DATA

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Types of Data Sets

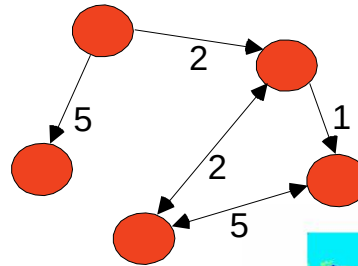
Record

- Relational records
- Data matrix, e.g., numerical matrix, crosstabs
- Document data: text documents: term-frequency vector
- Transaction data

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

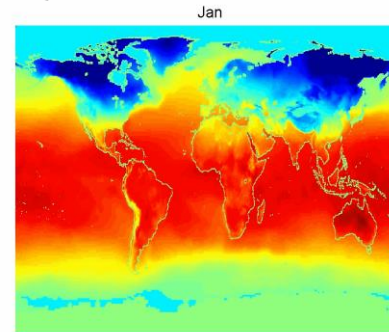
Graph and network

- World Wide Web
- Social or information networks
- Molecular Structures



Ordered

- Video data: sequence of images
- Temporal data: time-series
- Sequential Data: transaction sequences
- Genetic sequence data



GGTTCCGCCTTCAGCCCCGCGCC
 CGCAGGGCCCCGCCCCGCGCCGTC
 GAGAAGGGCCCCGCCTGGCGGGCG
 GGGGGAGGCGGGGCCGCCGAGC
 CCAACCGAGTCCGACCAGGTGCC
 CCCTCTGCTCGGCCTAGACCTGA
 GCTCATTAGGCGGCAGCGGACAG
 GCCAAGTAGAACACGCGAAGCGC
 TGGGCTGCCTGCTGCGACCAGGG

Spatial, image and multimedia

- Spatial data: maps
- Image data
- Video data

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Getting Familiar with Your Data

□ Knowledge about your data is useful for data preprocessing

- What are the types of attributes or fields that make up your data?
- What kind of values does each attribute have?
- Which attributes are discrete, and which are continuous-valued?
- What do the data look like?
- How are the values distributed?
- Are there ways we can visualize the data to get a better sense of it all?
- Can we spot any outliers?
- Can we measure the similarity of some data objects with respect to others?

Data Objects

- ❑ Data sets are made up of **data objects** – A data object represents an entity.
 - *e.g. in a university database, the objects maybe students, professors, and courses.*
- ❑ Also called:
 - *samples, examples, instances, data points, objects, tuples*
- ❑ They are described by attributes
- ❑ In a database:
 - **rows => data objects and columns => attributes**

What Is an Attribute?

- ❑ An **attribute** is a data field, representing a characteristic or feature of a data object.
 - *e.g. attributes describing a customer object can include, for example, customer ID, name, and address.*
- ❑ *Dimension, feature, and variable* are also used in place of attribute
- ❑ A set of attributes used to describe a given object is called an **attribute vector** (or feature vector).
 - *One attribute: univariate*
 - *Two attributes: bivariate*

Attribute Types

1) Nominal Attributes

2) Binary Attributes

3) Ordinal Attributes

4) Numeric Attributes :

➤ *Interval-scaled Attributes*

➤ *Ratio-scaled Attributes*

qualitative

quantitative

1. Attribute Type: Nominal Attributes

- ❑ Nominal means “relating to names.”
- ❑ The values of a **nominal attribute** are symbols or names of things.
- ❑ Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as *categorical*.
- ❑ The values do not have any meaningful order.
- ❑ It is possible to represent such symbols or “names” with numbers but the numbers are not intended to be used quantitatively.
 - *i.e. It makes no sense to subtract one customer ID number from another*

Nominal Attributes: Example

- ❑ Suppose that hair color and marital status are two attributes describing person objects.
 - *Hair color*: black, brown, blond, red, auburn, gray, and white.
 - *Marital status*: single, married, divorced, and widowed.
- ❑ Occupation, ID numbers, Zip codes etc... are nominal attributes, too.

2. Attribute Type: Binary Attributes

- ❑ A **binary attribute** is a nominal attribute with only two categories or states: 0 or 1,
 - **0** typically means that the *attribute is absent*
 - **1** means that the *attribute is present*
- ❑ Binary attributes are referred to as **Boolean** if the two states correspond to **true** and **false**.

Binary Attributes: Example

- ❑ Given the attribute *smoker* describing a patient object:
 - **1** indicates that the patient smokes
 - **0** indicates that the patient does not

- ❑ The patient undergoes a medical test that has two possible outcomes. The attribute *medical test* is binary:
 - **1** means the result of the test for the patient is positive
 - **0** means the result is negative.

Symmetric and Asymmetric Binary Attributes

- ❑ A binary attribute is **symmetric** if the outcomes of the states are equally important;
 - e.g. the **attribute gender** having the states male and female.
 - **There is no preference** on which outcome should be coded as 0 or 1.

- ❑ A binary attribute is **asymmetric** if the outcomes of the states are not equally important;
 - e.g. the positive and negative outcomes of a **medical test for HIV**.
 - **Convention: assign 1 to most important outcome** (e.g., HIV positive).

3. Attribute Type: Ordinal Attributes

❑ An **ordinal attribute** is an attribute with possible values that have a **meaningful order** or **ranking** among them, but the **magnitude between successive values is not known**.

➤ e.g. *drink size* at a fast-food restaurant can be an ordinal attribute with values: *small, medium, large*.

❑ Ordinal attributes may also be obtained from the discretization of numeric quantities by splitting the value range into a finite number of ordered categories.

➤ e.g. *age* info can be used as an ordinal attribute: ages between 0-13 can be taken as *child*, ages between 13-18 as *young*, and 18-above as *adult*.

Ordinal Attributes: Example

□ Grade can be an ordinal attribute:

➤ A+, A, A-, B+, and so on...

□ Professional ranks can be enumerated in a sequential order:

➤ assistant, associate, and full for professors

□ Army Ranks:

➤ private, private first class, specialist, corporal, and sergeant.

□ Ordinal attributes are often used in surveys for ratings (*for subjective assessments of qualities that cannot be measured objectively*).

➤ Ask a customer to rate how satisfied they were. Customer satisfaction had the following ordinal categories;

- 0: very dissatisfied
- 1: somewhat dissatisfied
- 2: neutral
- 3: satisfied
- 4: very satisfied

NOTE - 1

- ❑ Note that nominal, binary, and ordinal attributes are **qualitative**.
 - *They describe a feature of an object without giving an actual size or quantity.*
- ❑ The values of such qualitative attributes are typically words representing categories. If integers are used, they represent computer codes for the categories, as opposed to measurable quantities (e.g., 0 for small drink size, 1 for medium, and 2 for large).
- *Now we are going to look at numeric attributes, which provide **quantitative** measurements of an object.*

4. Attribute Type: Numeric Attributes

- ❑ A **numeric attribute** is *quantitative*; that is, it is a *measurable quantity*, represented in integer or real values.
- ❑ Numeric attributes can be interval-scaled or ratio-scaled.
 - **Interval-scaled attributes** are measured on a scale of equal-size units.
 - The values have order and can be positive, 0, or negative.
 - They allow us to compare and quantify the difference between values
 - e.g. Temperatures in Celsius and Fahrenheit, Calendar years
 - A **ratio-scaled attribute** is a numeric attribute with an inherent zero-point.
 - We can speak of a value as being a multiple (or ratio) of another value (10 K° is twice as high as 5 K°).
 - The values have order and difference between values can be calculated
 - e.g. Temperature in Kelvin, years of experience, number of words

Discrete versus Continuous Attributes

- ❑ A **discrete attribute** has a finite or countably infinite set of values, which may or may not be represented as integers.
 - The attributes *hair color*, *smoker*, *medical test*, and *drink size* each have a finite number of values, and so are discrete.
 - An attribute is countably infinite if the set of possible values is infinite but the values can be put in a one-to-one correspondence with natural numbers. For example, the attributes *customer ID* and *zip codes* are countably infinite.

Discrete versus Continuous Attributes

- ❑ If an attribute is not discrete, it is **continuous**.
- ❑ The terms *numeric attribute* and *continuous attribute* are often used interchangeably in the literature.
 - *This can be confusing because, in the classic sense, continuous values are real numbers, whereas numeric values can be either integers or real numbers.*
 - *In practice, real values are represented using a finite number of digits. Continuous attributes are typically represented as floating-point variables.*

Summary of Attributes

Attribute Type	Description	Examples
Nominal	Nominal attribute values only distinguish. ($=$, $\neq$)	zip codes, employee ID, eye color, sex: {male, female}
Ordinal	Ordinal attribute values also order objects. ($<$, $>$)	hardness of minerals, {good, better, best}, grades, street numbers
Interval	For interval scaled attributes, differences between values are meaningful. ($+$, $-$)	calendar dates, temperature in Celsius or Fahrenheit
Ratio	For ratio scaled variables, both differences and ratios are meaningful. ($*$, $/$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current

Basic Statistical Descriptions of Data

- For data preprocessing to be successful, it is essential to have an overall picture of your data.
 - i. **Measures of central tendency:** Given an attribute, where do most of its values fall? Calculate the mean, median, mode, and midrange.
 - ii. **Dispersion of the data:** How are the data spread out? Most common ones: range, quartiles, and interquartile range; the five-number summary and boxplots; and the variance and standard deviation of the data.
 - iii. **Graphic displays** of basic statistical descriptions to **visually inspect your data**. Bar charts, pie charts, line graphs, quantile plots, quantile–quantile plots, histograms, and scatter plots.

I. Measuring the Central Tendency

□ Measures of central tendency include:

- Mean
- Median
- Mode
- Midrange

Mean

- The most common and effective numeric measure of the “center” of a set of data is the **(arithmetic) mean**.
- Let $x_1, x_2, \dots, x_N$ be a set of N values or observations, such as for some numeric attribute X , like salary. The mean of this set of values is:

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

Mean

- When each value x_i in a set is associated with a weight w_i for $i=1,\dots,N$. (The weights reflect the significance, importance, or occurrence frequency attached to their respective values.)

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + \dots + w_N}$$

- This is called the **weighted arithmetic mean** or the **weighted average**.

Mean

- ❑ Mean is sensitive to extreme (outlier) values so it is not always the best way for measuring the center of the data.
 - *e.g. the mean score of a class in an exam could be pulled down quite a bit by a few very low scores.*
- ❑ **Trimmed mean**, which is the mean obtained after chopping off values at the high and low extremes can be used in this case.
 - *e.g. first sort the values observed then remove the top and bottom 2%*
 - *Avoid trimming too large (%20) a portion, as this can result in the loss of valuable information*

Median

- ❑ For skewed (asymmetric) data, a better measure of the center of data is the **median**, which is the middle value in a set of ordered data values.
- ❑ It is the value that separates the higher half of a data set from the lower half.
- ❑ In probability and statistics, the median generally applies to numeric data; however, we may extend the concept to ordinal data, too.

Median

□ Suppose that a given data set of N values for an attribute X is sorted in increasing order.

➤ *If N is odd*, then the median is the middle value of the ordered set.

➤ *If N is even*, then the median is not unique; it is the two middlemost values and any value in between. If X is a numeric attribute in this case, by convention, the median is taken as the average of the two middlemost values.

Median

- If data are grouped in intervals according to their x_i data values and that the frequency (i.e., number of data values) of each interval is known.

$$median = L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width$$

- Interval that contains the median frequency is median interval:
 - L_1 is the lower boundary of the median interval
 - N is the number of values in the entire data set,
 - $(\sum freq)_l$ is the sum of the frequencies of all of the intervals that are lower than the median interval
 - $freq_{median}$ is the frequency of the median interval
 - $width$ is the width of the median interval

Example

Age	Frequency
1-5	200
5-15	450
15-20	300
20-50	1500
50-80	700
80-110	44

- Calculate the median according to the given information:

$$median = L_1 + \left(\frac{N/2 - (\sum freq)_l}{freq_{median}} \right) width$$

$$median = 20 + \left(\frac{(3194/2) - (200 + 450 + 300)}{1500} \right) \times 30$$
$$= 32.94$$

Mode

- ❑ The **mode** for a set of data is the value that occurs most frequently in the set.
- ❑ Therefore, it can be determined for qualitative and quantitative attributes.
- ❑ Data sets with one, two, or three modes are respectively called *unimodal*, *bimodal*, and *trimodal*.
- ❑ In general, a data set with two or more modes is *multimodal*.
- ❑ At the other extreme, if each data value occurs only once, then there is *no mode*.

Midrange

- ❑ The **midrange** can also be used to assess the central tendency of a numeric data set.
- ❑ It is the average of the largest and smallest values in the set.

Example

- Salary values (in thousands of dollars) of a company are given as below:

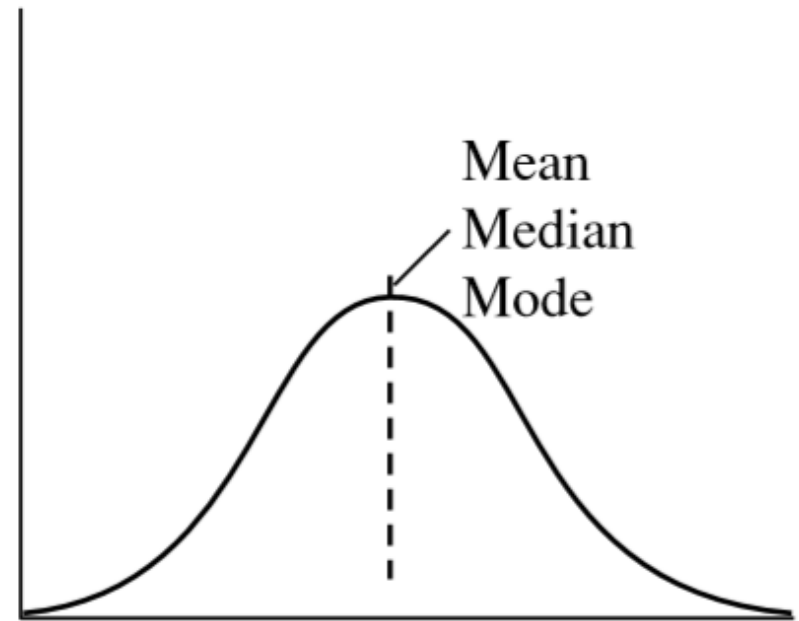
56, 30, 110, 36, 47, 50, 63, 52, 70, 60, 70, 52

Find the mean, median, mode, and midrange of the data...

- ✓ Mean: \$58,000
- ✓ Median: \$54,000
- ✓ Mode: Bimodal, the two modes are \$52,000 and \$70,000.
- ✓ Midrange: \$70,000

NOTE - 2

- ❑ In a unimodal frequency curve with perfect **symmetric data** distribution, the mean, median, and mode are all at the same center value.



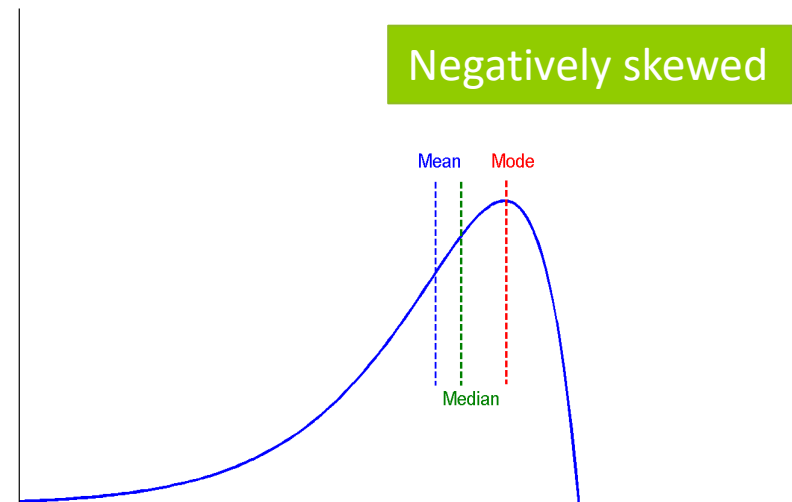
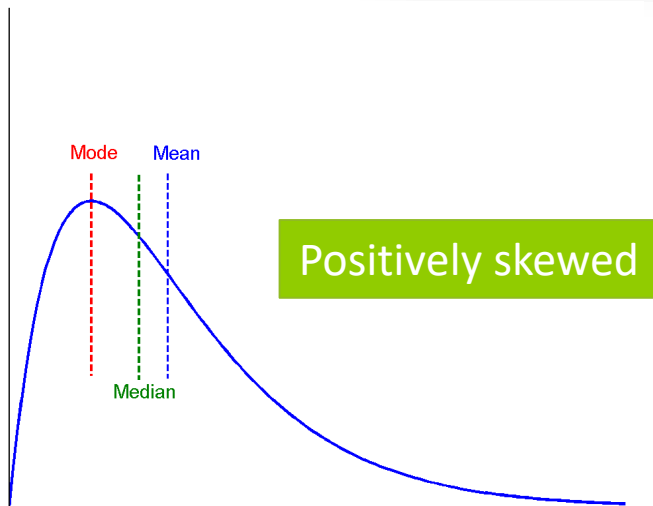
(a) Symmetric data

NOTE - 3

❑ Data in most real applications are not symmetric.

- **Positively skewed**: Mode occurs at a value that is smaller than the median
- **Negatively skewed**: mode occurs at a value greater than the median.

$$mean - mode \approx 3 \times (mean - median)$$



NOTE - 4

- ❑ The central tendency of an ordinal attribute can be represented by its mode and its median, but the mean cannot be defined.
- ❑ Interval-scaled and ratio-scaled attributes are numeric, so we can compute their mean value, in addition to the median and mode measures of central tendency.

II. Measuring the Dispersion of Data

□ Measures to assess the dispersion or spread of numeric data:

- Range
- Quantiles
- Quartiles
- Percentiles
- Interquartile range
- Five-number summary
- Variance and standard deviation

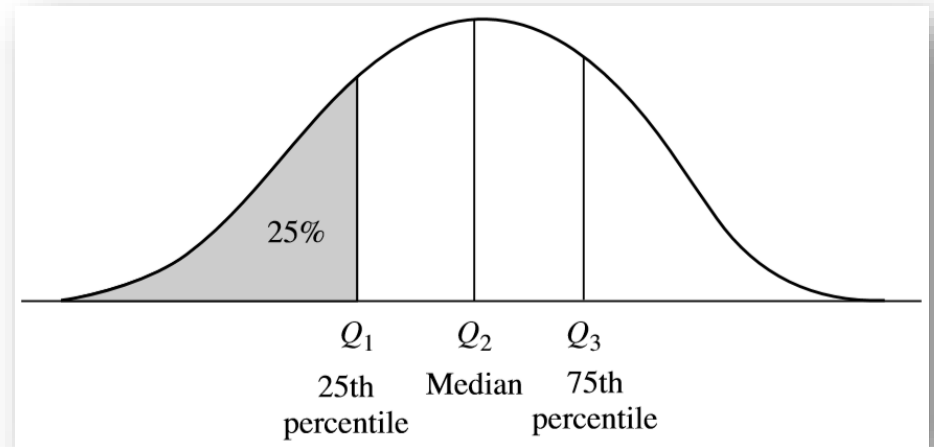
Range and Quantiles

Let $x_1, x_2, \dots, x_N$ be a set of observations for some **numeric attribute**, X .

- ❑ The **range** of the set is the ***difference between the largest ($\max()$) and smallest ($\min()$)*** values.
- ❑ Suppose that the data for attribute X are sorted in increasing numeric order. If we can pick certain data points so as to split the data distribution into **equal-size consecutive sets**. These data points are called **quantiles**.

Quartiles and Percentiles

- ❑ The **2-quantile** is the data point dividing the lower and upper halves of the data distribution. It corresponds to the **median**.
- ❑ The **4-quantiles** are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as **quartiles**.
- ❑ The **100-quantiles** are more commonly referred to as **percentiles**; they divide the data distribution into 100 equal-sized consecutive sets.



Interquartile Range

- ❑ The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data.
- ❑ This distance is called the **interquartile range (IQR)** and is defined as

$$IQR = Q_3 - Q_1$$

Example:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

The quartiles for this data are:

$Q_1 = (47+50)/2 = 48.5$ $Q_2 = (52+56)/2 = 54$ $Q_3 = (63+70)/2 = 66.5$.

Thus, the interquartile range is $IQR = 66.5 - 48.5 = \$18,000$.

Five-Number Summary, Boxplots, and Outliers

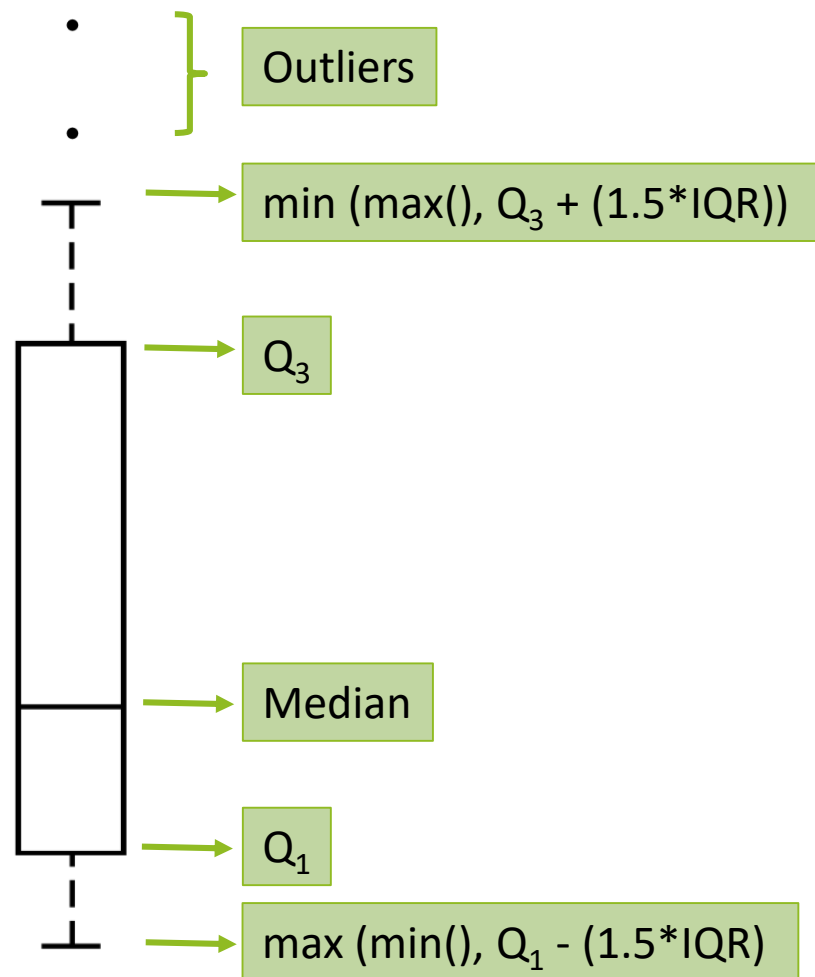
- ❑ No single numeric measure of spread (e.g., IQR) is very useful for describing skewed distributions.
- ❑ Therefore, it is more informative to also provide the two quartiles Q_1 and Q_3 , along with the median.
- ❑ A common rule of thumb for **identifying suspected outliers** is to *single out values falling at least $1.5 \times IQR$ above the third quartile or below the first quartile.*

Five-Number Summary, Boxplots, and Outliers

❑ The **five-number summary** of a distribution consists of the median (Q_2), the quartiles Q_1 and Q_3 , and the smallest and largest individual observations, written in the order of **Minimum, Q_1 , Median, Q_3 , Maximum**.

❑ Boxplots are a popular way of visualizing a distribution.

❑ Potential outliers are shown individually.



Variance and Standard Deviation

□ **Variance** and **standard deviation** indicate how spread out a data distribution is.

- A *low standard deviation* means that the data observations tend to be very close to the mean
- A *high standard deviation* indicates that the data are spread out over a large range of values.

Variance and Standard Deviation

□ The **variance** of N observations, $x_1, x_2, \dots, x_N$, for a numeric attribute X :

➤ $\bar{x}$: mean value of the observations

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2$$

□ The **standard deviation**, σ , of the observations is the square root of the variance, σ^2 .

Example

❑ Values were: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

❑ We found $\bar{x} = \$58,000$

❑ $N = 12$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2$$

❑ So variance and standard deviation can be calculated as;

$$\sigma^2 = \frac{1}{12} (30^2 + 36^2 + 47^2 \dots + 110^2) - 58^2$$

$$\approx 379.17$$

$$\sigma \approx \sqrt{379.17} \approx 19.47.$$

III. Graphic Displays

□ Graphic displays of basic statistical descriptions are:

- Quantile Plots
- Quantile–Quantile Plots
- Histograms
- Scatterplots

□ The first three of these show *univariate distributions* (i.e., data for one attribute), while scatter plots show *bivariate distributions* (i.e., involving two attributes).

DATA



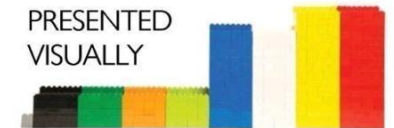
SORTED



ARRANGED



PRESENTED
VISUALLY



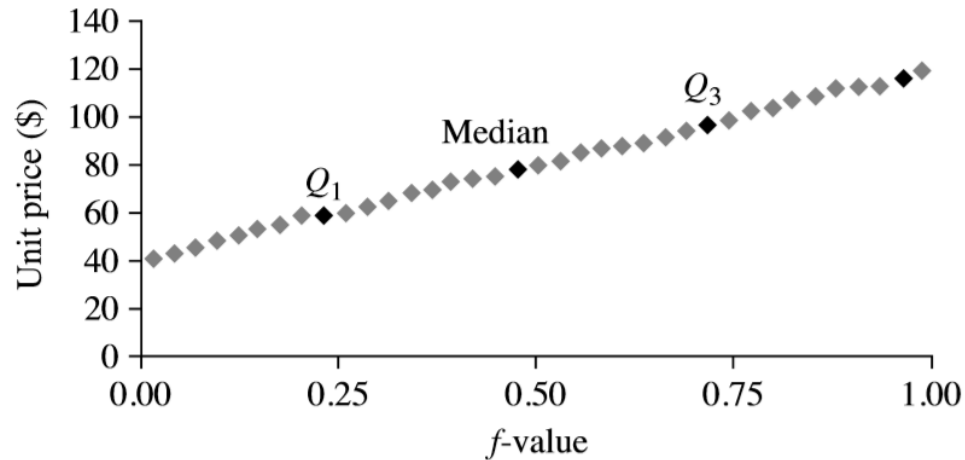
EXPLAINED
WITH A STORY



Quantile Plot

❑ **Quantile plot** displays all of the data for the given attribute (allowing the user to assess both the overall behavior and unusual occurrences)

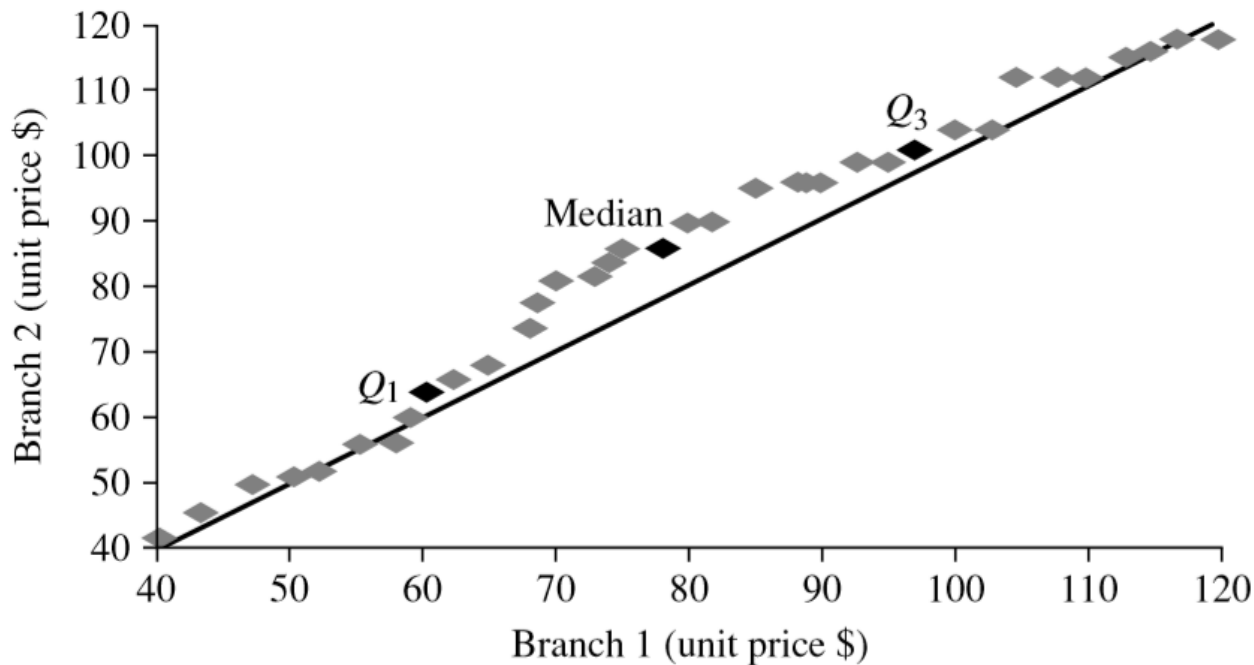
❑ Second, it plots quantile information



Unit price (\$)	Count of items sold
40	275
43	300
47	250
—	—
74	360
75	515
78	540
—	—
115	320
117	270
120	350

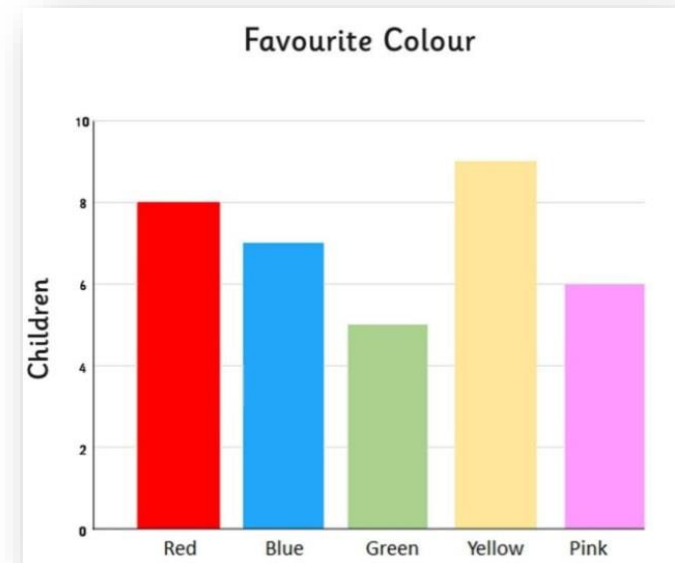
Quantile–Quantile Plot

□ A **quantile–quantile plot**, or **q-q plot**, graphs the quantiles of one univariate distribution against the corresponding quantiles of another.



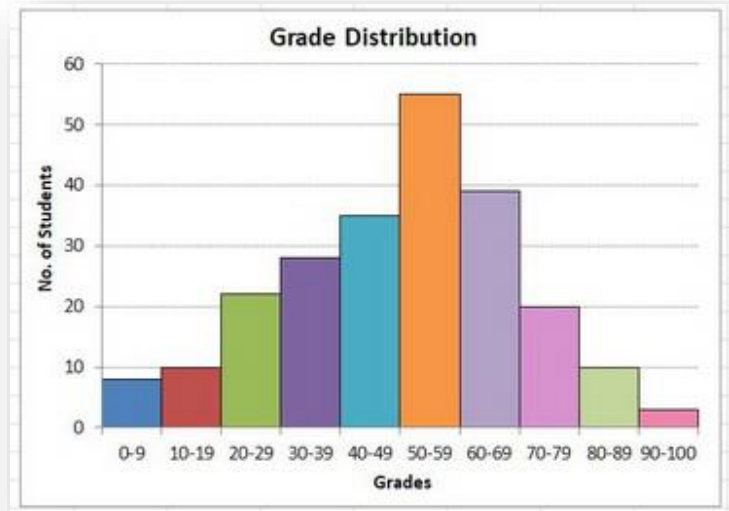
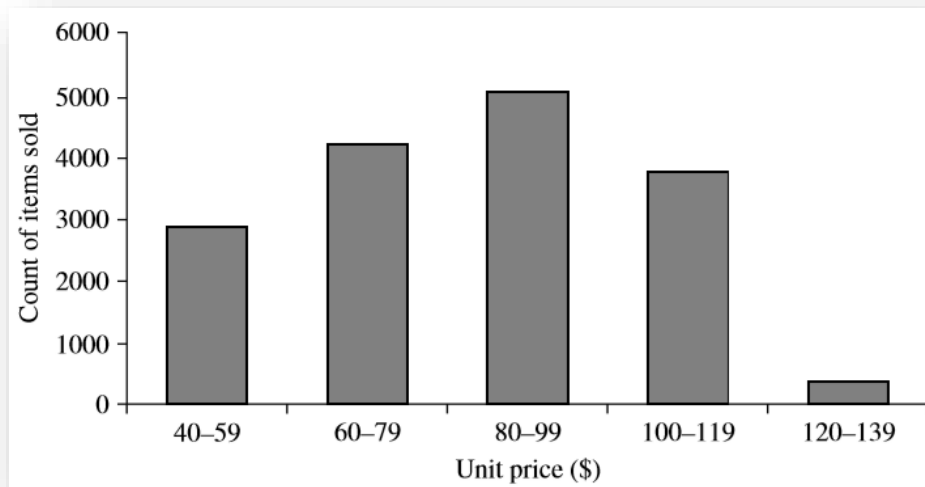
Histograms

- ❑ **Histograms** (or frequency histograms) are at least a century old and are widely used.
- ❑ Plotting histograms is a graphical method for summarizing the distribution of a given attribute, X .
- ❑ If X is nominal, such as *automobile_model* or *item_type*, then a pole or vertical bar is drawn for each known value of X . The height of the bar indicates the frequency(i.e.,count) of that X value. The resulting graph is more commonly known as a *bar chart*.

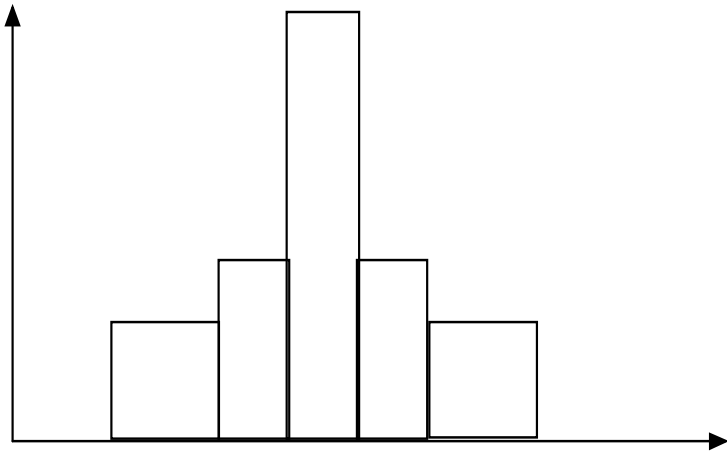
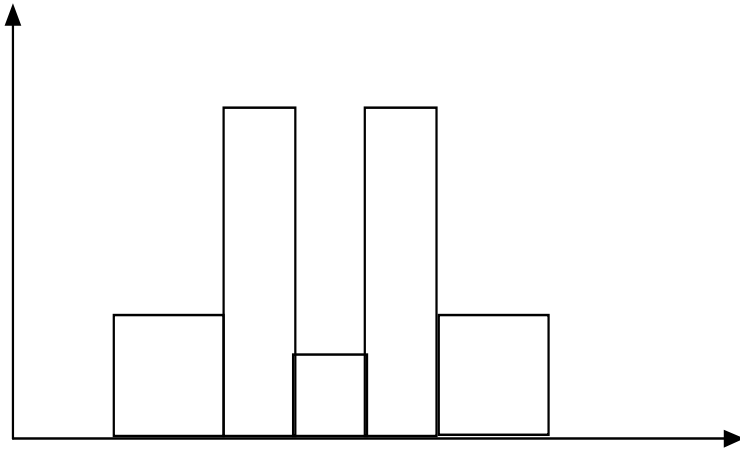


Histograms

- ❑ If X is numeric, the term histogram is preferred.
- ❑ The range of values for X is partitioned into disjoint consecutive subranges (buckets or bins).
- ❑ The range of a bucket is known as the width. Typically, the buckets are of equal width.



Histograms Often Tell More than Boxplots



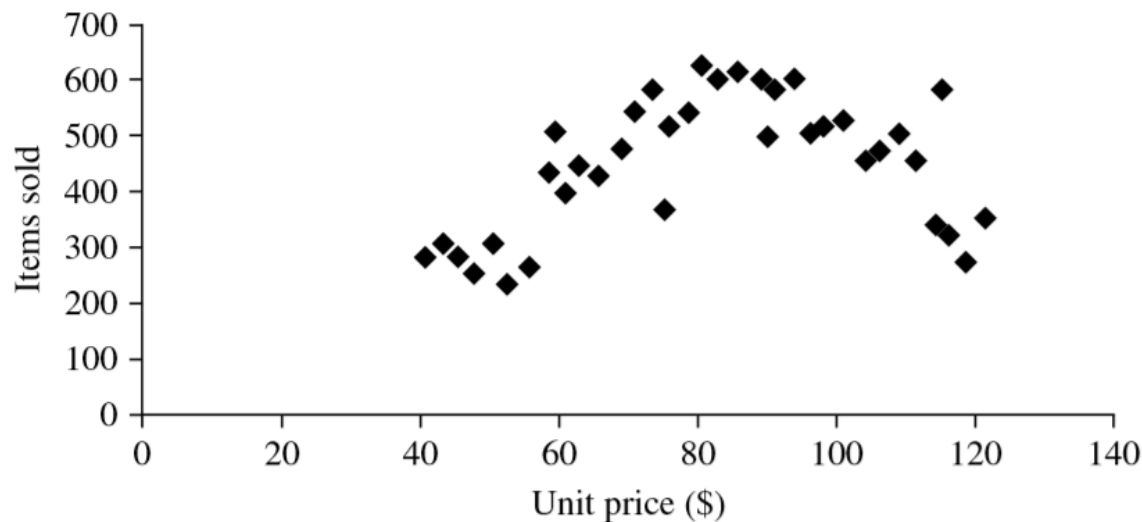
❑ The two histograms shown in the left may have the same boxplot representation

➤ The same values for: min, Q1, median, Q3, max

❑ But they have rather different data distributions

Scatter Plots and Data Correlation

- ❑ A **scatter plot** is one of the most effective graphical methods for determining if there appears to be a **relationship, pattern, or trend between two numeric attributes**.
- ❑ To construct a scatter plot, each pair of values is treated as a pair of coordinates in an algebraic sense and plotted as points in the plane.



Scatter Plots and Data Correlation

- Used to see clusters of points and outliers, or to explore the possibility of correlation relationships

